

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm Examination

November 2016

PH211 Pharmaceutical Analysis – II

Date: 08.11.2016, Tuesday Time: 1:30p.m to 4:30p.m Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the following

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|---|-----|--|----|
| A | i) | Write the advantages and disadvantages of Instrumental methods of analysis. Classify the Instrumental methods of analysis | 08 |
| | ii) | Explain the following terms:
a) Equivalent conductance b) Specific conductance | 02 |
| B | i) | Explain Kohlrausch law. Write a note on factors affecting conductance. | 05 |
| | ii) | Discuss the principle of Polarography and explain typical current voltage curve. | 05 |
| C | i) | Comment on following:
a) Equivalent conductance increases on dilution of electrolytic solution.
b) Equivalent conductance decrease with increase in concentration. | 05 |
| | ii) | Explain the working of DME and write the advantages and limitations of DME. | 05 |

Q 2 Attempt any **TWO** of the following

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|---|------|---|----|
| A | i) | What is S/N ratio? Explain the source of noise in instrumental analysis. | 04 |
| | ii) | Enlist characteristics of an ideal reference electrode used in potentiometry. Give a brief account of various reference electrodes. | 06 |
| B | i) | Write an explanatory note on pulse polarography. | 05 |
| | ii) | Write a short-note on Biamperometry titrations | 05 |
| C | i) | Write the factors affecting limiting current and diffusion current. | 05 |
| | iii) | Give a detailed account of Glass Electrode | 05 |

SECTION- II

Q 3	Attempt any <u>TWO</u> of the following	A	[1] Write short note on Thermo Gravimetric Analysis.	05
		A	[2] What are the significant characteristics of a thermobalance used in TGA?	05
		B	Write a note on principle of polarimetry technique. Explain the instrumentation of polarimetry with the help of suitable diagram.	10
		C	1) Explain the theories involved in Chromatography.	05
			2) Give the classification of chromatographic techniques.	05
Q 4	Attempt any <u>TWO</u> of the following	A	Describe the steps involved in the analysis of a substance by thin layer chromatography	10
		B	Define:	10
			1) Chromatography,	
			2) Optically active compound.	
			3) Retention time,	
			4) HETP,	
		C	5) Tailing Factor	05
			1) Write a note on instrumentation involved in the Differential Scanning Calorimetric Technique.	05
			2) Write a note on Van Deemter Equation.	05